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Batch: B16

Program: Btech CSE

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Experiment -1
Linux Programming Environment & Basic Commands

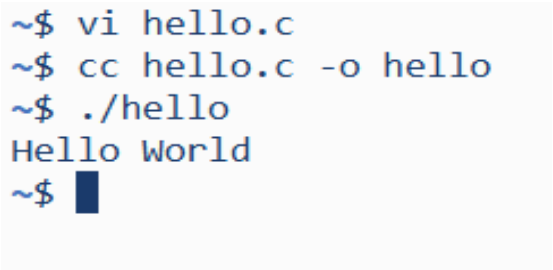
A) Setting up of Linux Programming Environment

We can do C programming online or offline according to the needs. It is required for us to keep working on a Linux Environment.

Online: Cocalc.ai –

1. Open CoCalc.ai and sign in.
2. Create/open a project and launch a Linux terminal.
3. Create a C source file (for example, hello.c).
4. Write and save the C program.
5. Compile the program using gcc.
6. Execute the program and observe the output.

CoCalc.ai project/terminal showing the Linux environment and your C program.



Offline:

Methods: Dual boot, Single boot, Bootable Pendrive, VM

Dual Boot – Two operating systems are installed on the same computer and the user selects one while starting the computer.

Single Boot – One operating system is installed and used as the main operating system.

Bootable Pendrive – A USB drive containing a bootable operating system is used to start the computer.

Virtual Machine (VM) – A virtual computer is created inside an existing operating system using virtualization software.

1. Virtual Machine Manager (Hypervisor)

Type-1 (Baremetal)

It is installed directly on the computer hardware. It does not require another operating system between the hardware and hypervisor.

Working:

Hardware → Hypervisor → Virtual Machines → Operating Systems

Examples: VMware ESXi, Microsoft Hyper-V Server, Xen.

Main point: It provides better performance and efficiency and is commonly used on servers and data centers.

Type-2 (Hosted)

It is installed on an existing operating system, just like normal software.

For example, if we have Windows, we can install Oracle VirtualBox on Windows and then install Kali Linux inside VirtualBox.

Working:

Hardware → Host Operating System → Hypervisor → Virtual Machine → Guest OS

Examples: Oracle VirtualBox, VMware Workstation, VMware Fusion.

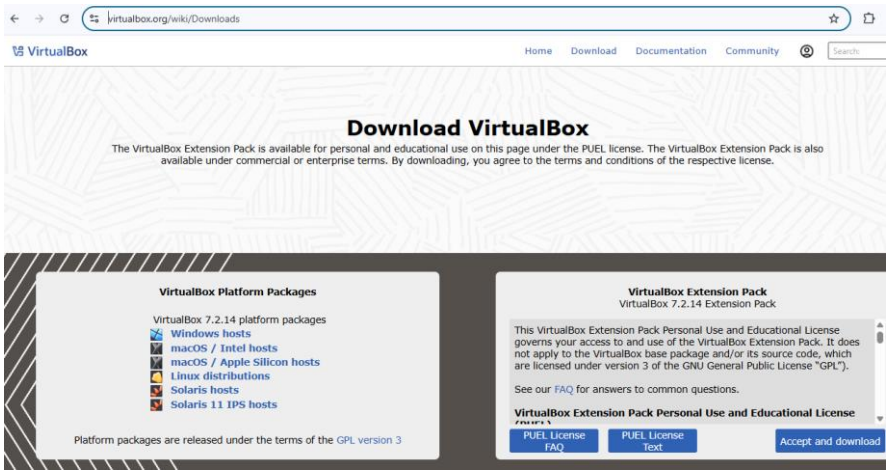
Main point: It is easy to install and use, so it is commonly used for learning, testing, and programming.

In Our C Programming Project.

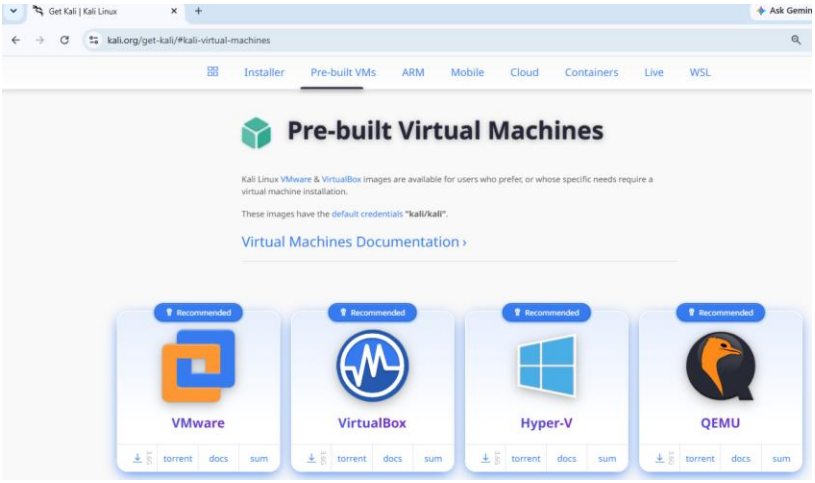
We are using Oracle VirtualBox, which is a Type 2 (Hosted) Hypervisor. We will install Kali Linux as a virtual machine inside VirtualBox and use it for C programming.

2. Installation of VMM & VM

Step-1: Download Oracle Virtual Box for “Windows hosts”

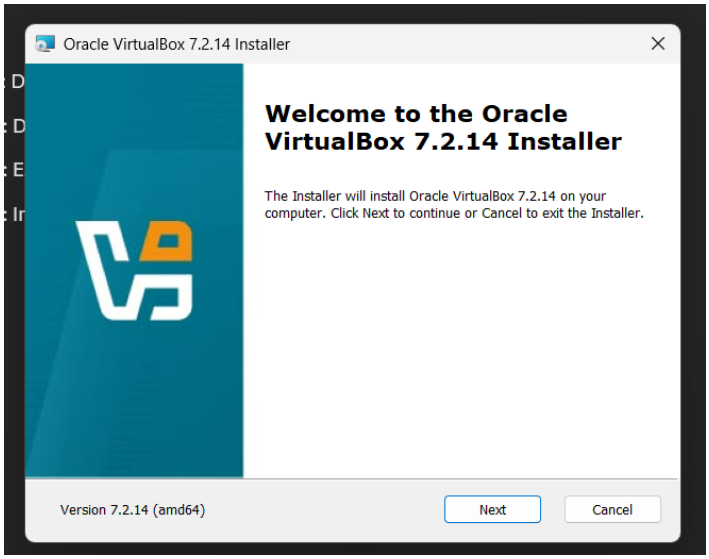


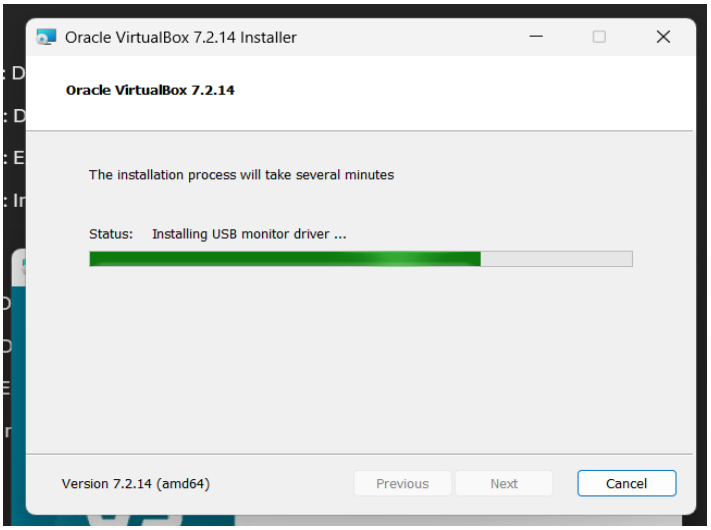
Step-2: Download Pre-built Kali Linux VM for Virtualbox



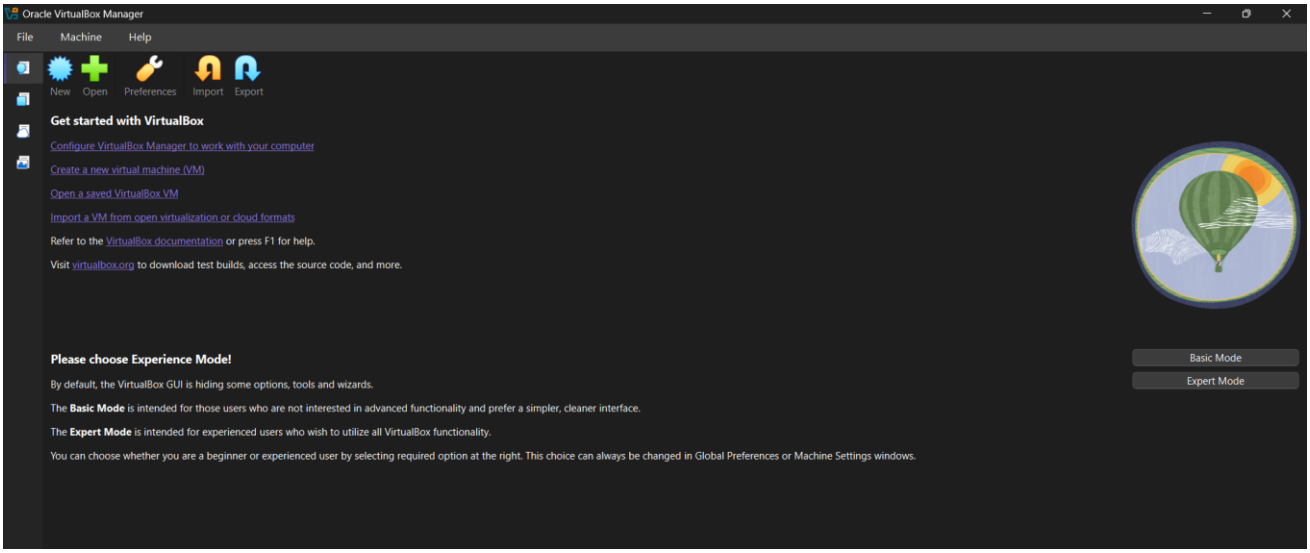
Step-3: Extracted the Kali Linux downloaded file

Step-4: Install Virtualbox

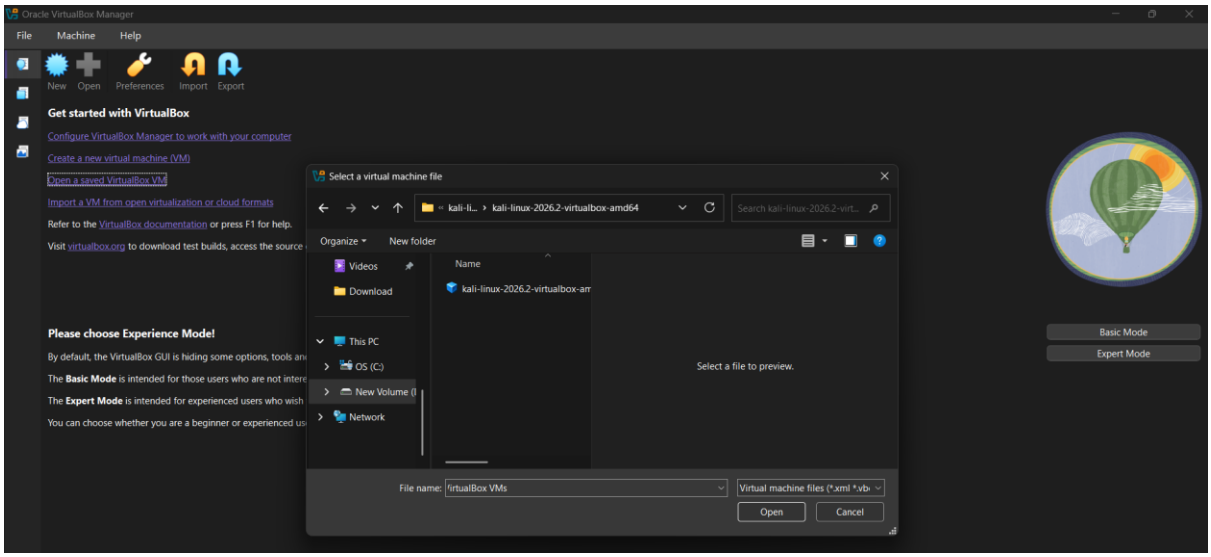


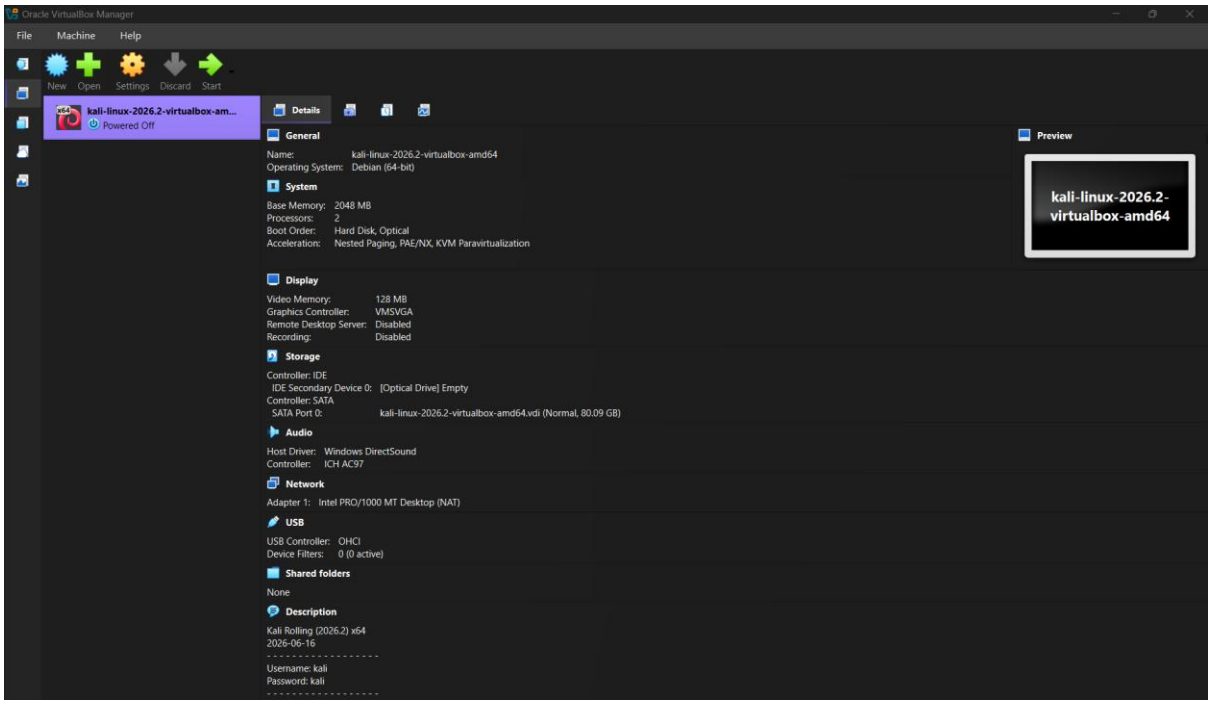


Step-5: Open Virtual box

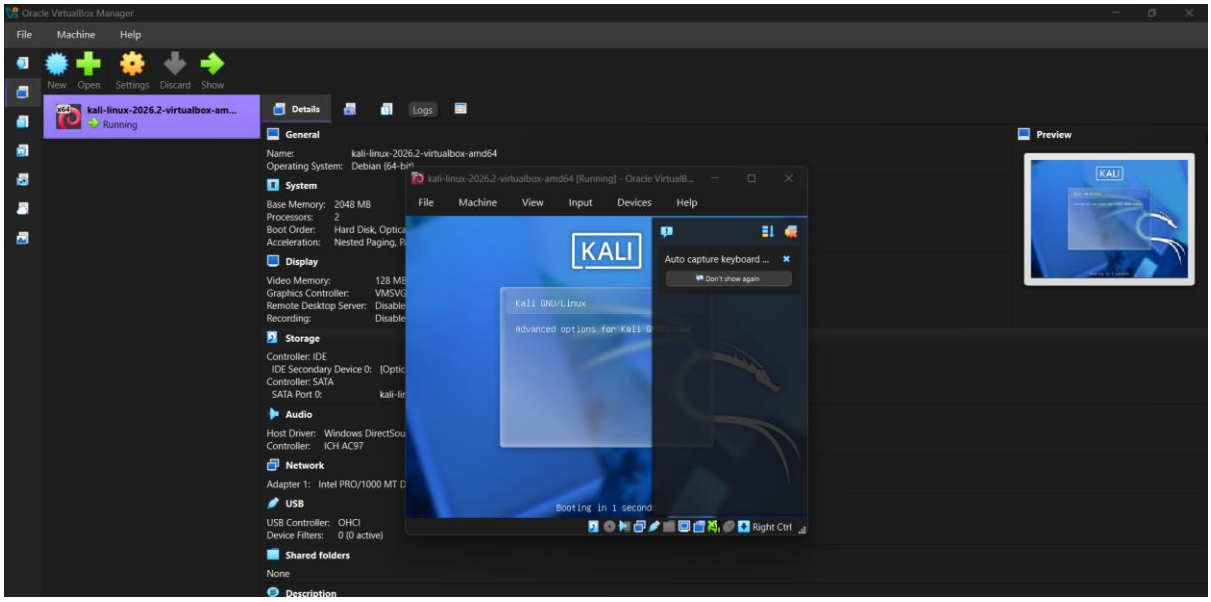


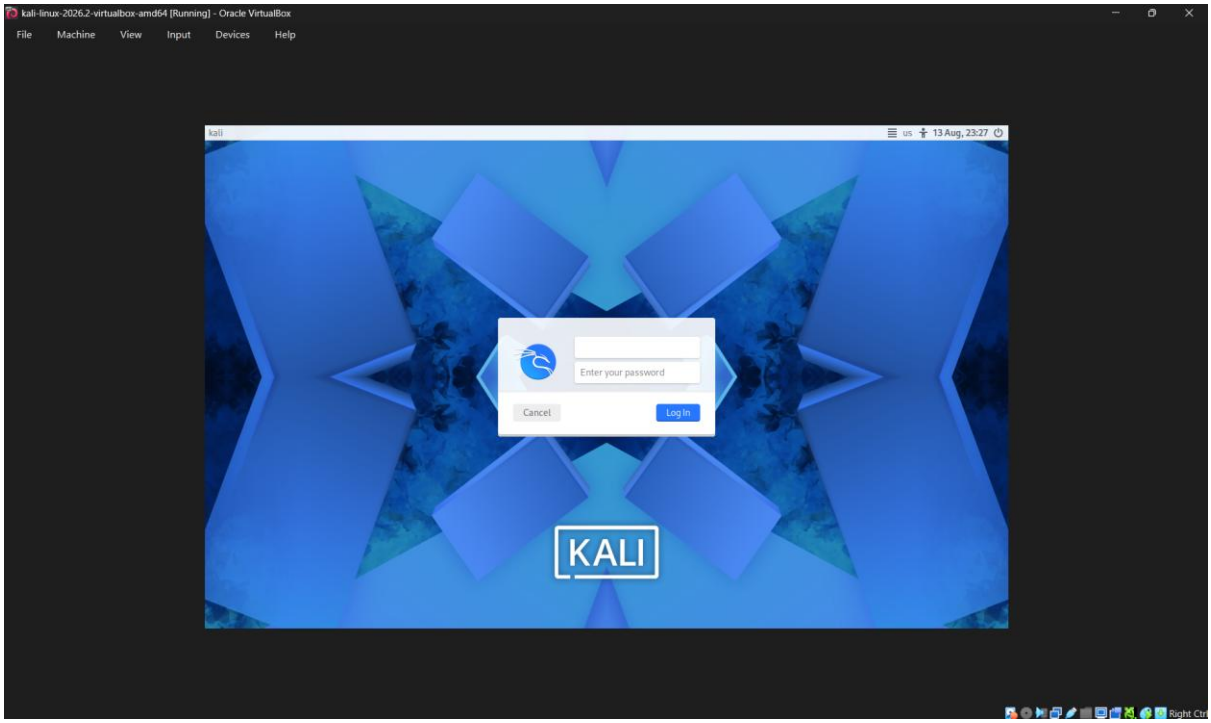
Step-6: Open the “saved VM”





Step-7: Start the VM





Thus, the installation of VMM (Hypervisor Type-2 Oracle VirtualBox) & VM (Kali Linux) is completed.

Step-8: Open the terminal and create the following simple program, compile and execute to understand the working of the environment.

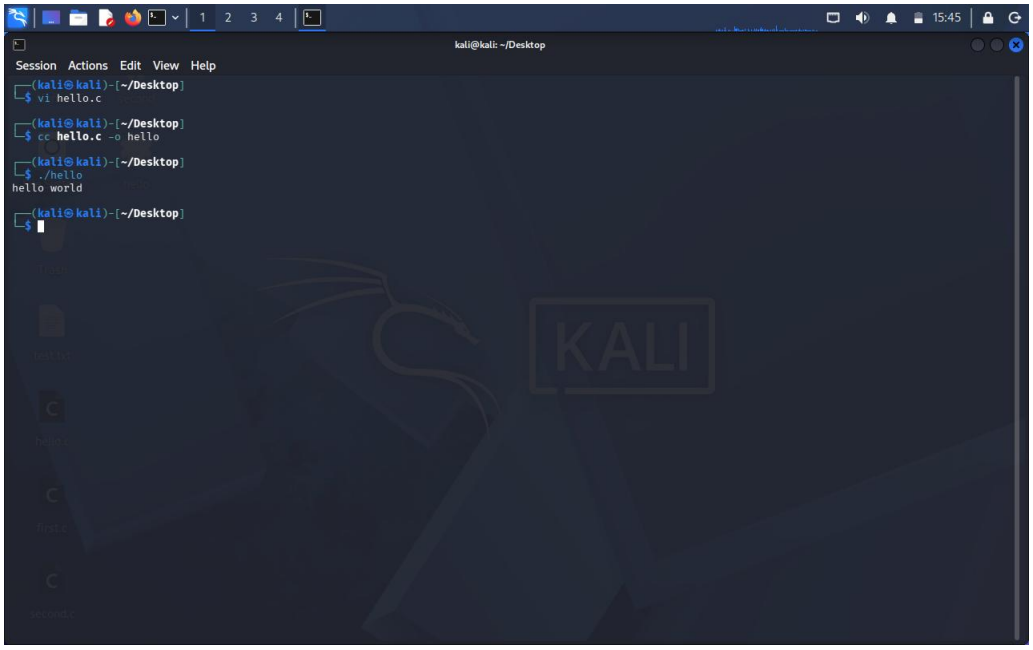
```
#include <stdio.h>

int main()
{
    printf("Hello World");
    return 0;
}
```

Compile command: gcc hello.c -o hello

Execution command: ./hello

Expected output: Hello World

Terminal showing the C program, compilation/execution and “Hello World” output.



B) **Working with Linux Commands**

Execute the following commands in your environment, take output snippets, and provide a brief explanation.

Terminal basics

pwd – Shows the present working directory.

mkdir – Creates a new directory.

rmdir – Removes an empty directory.

dir – Lists files and directories.

ls – Lists files and directories.

cd – Changes the current directory.

cd .. – Moves to the parent directory.

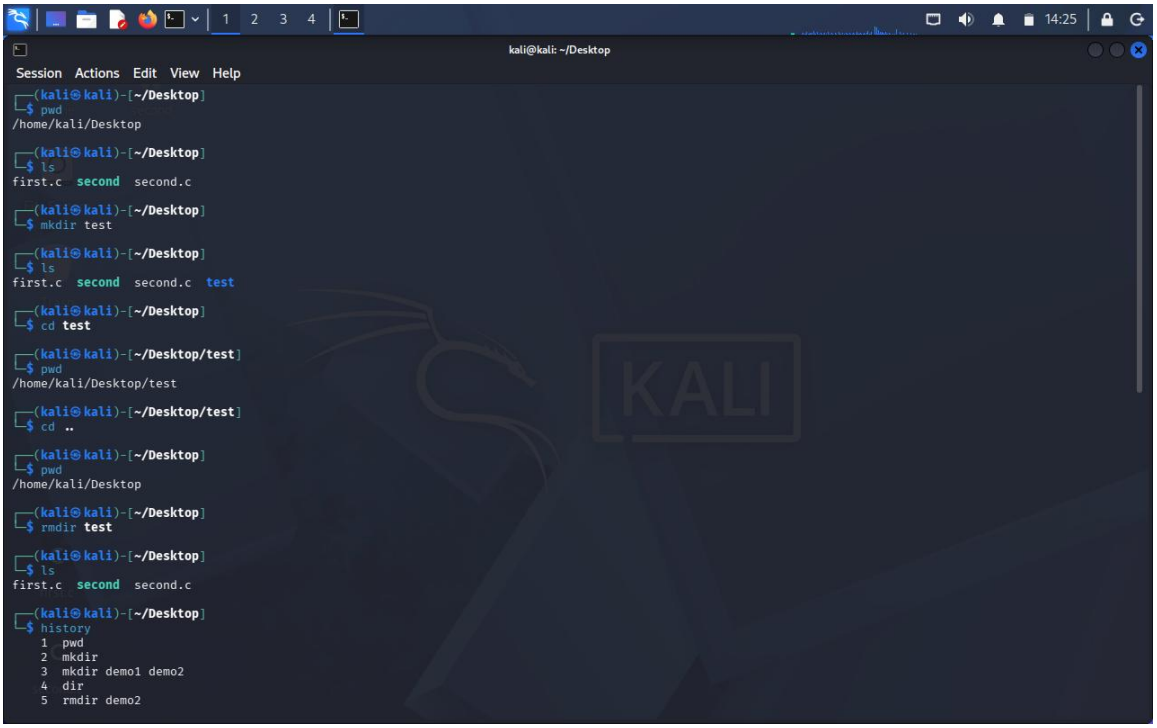
clear – Clears the terminal screen.

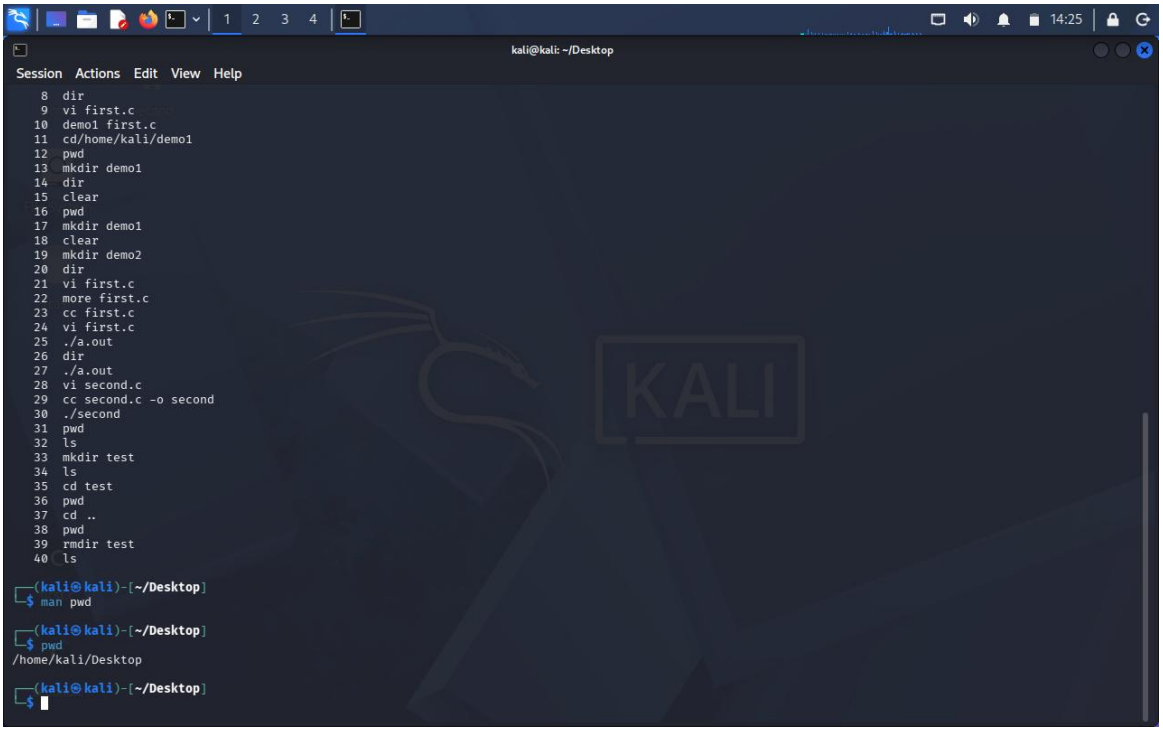
history – Shows previously executed commands.

man – Shows the manual/help page of a command.

whoami – Shows the current username.

The terminal output screenshot for the terminal basics commands.





Files

vi – Opens the vi text editor.

file – Shows information about a file type.

rm – Removes files/directories.

cp – Copies files/directories.

mv – Moves or renames files/directories.

cat – Displays file contents.

more – Displays contents page by page.

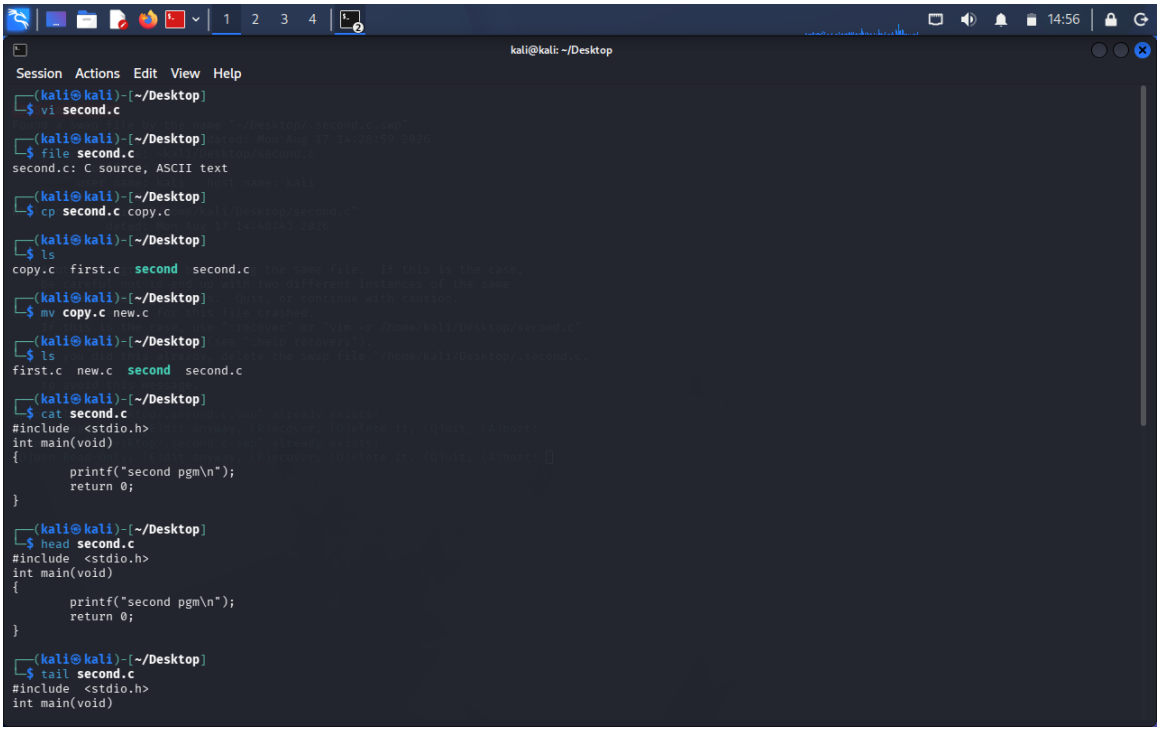
less – Views file contents with navigation.

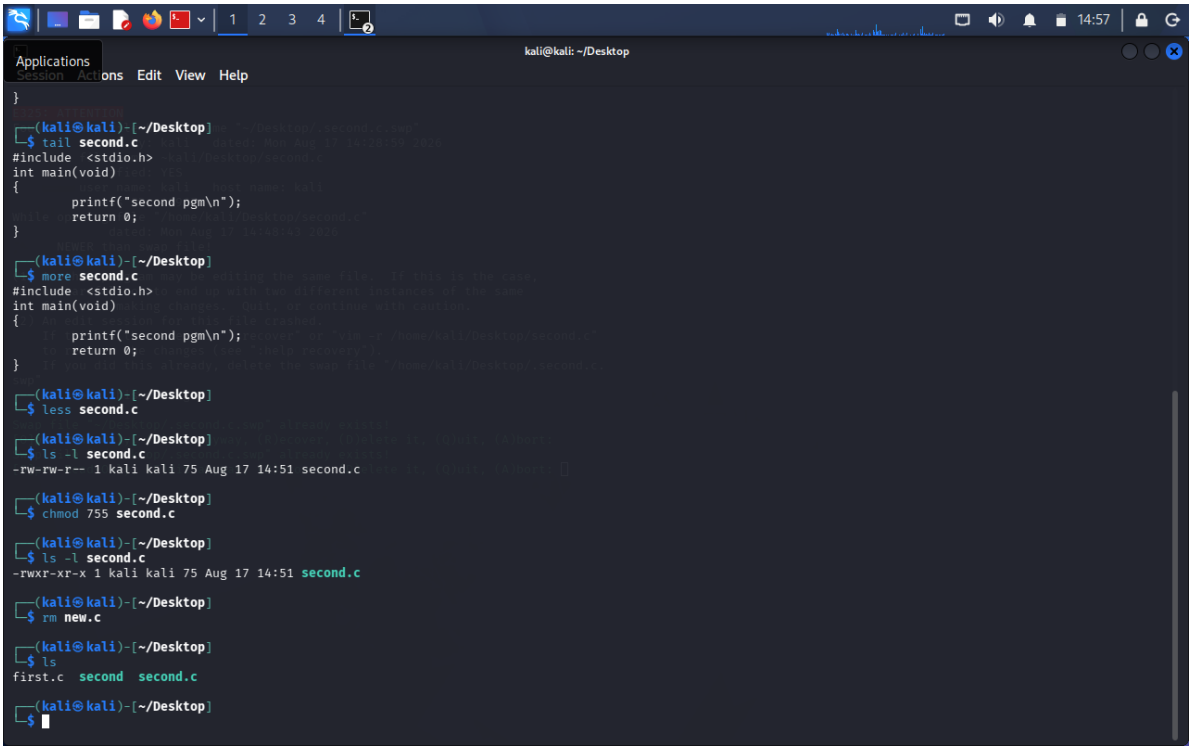
head – Displays the beginning of a file.

tail – Displays the end of a file.

chmod – Changes file permissions.

The terminal output screenshot for the files commands.





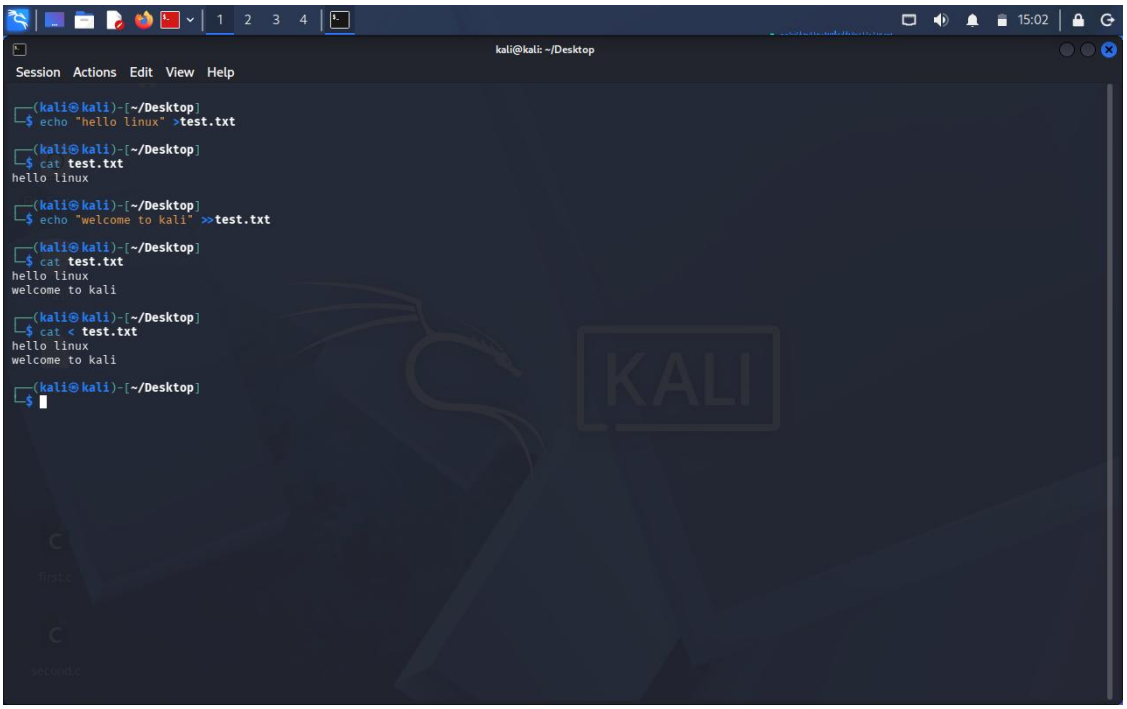
Redirection

> – Redirects output to a file, replacing its contents.

>> – Redirects output and appends it to a file.

< – Takes input from a file.

The terminal output screenshot for the redirection commands.



Result

The Linux programming environment was set up and the basic Linux commands were executed successfully